# Sharath Naganna Honnavalli

2nd Year MS (Research), Computer Science and Engineering, IIT Bombay
sharathhn@cse.iitb.ac.in | Website | GitHub | LinkedIn

## Research Interests

Mechanistic interpretability, focused on understanding and controlling how models represent concepts internally. Particularly interested in whether the same concept shares an internal structure across notations, prompting regimes, and modalities.

## Education

**Indian Institute of Technology Bombay** — Maharashtra, India
*M.S. (by Research), Computer Science and Engineering* | *GPA 9.48/10* — 2024 – Dec 2026 (expected)
Advisors: Dr. Pushpak Bhattacharyya; Dr. Swaprava Nath; Dr. Aditya Joshi (UNSW)

**JSS Science and Technology University** — Karnataka, India
*B.E., Computer Science and Engineering* | *GPA 9.51/10* — 2018 – 2022

## Publications

1. **Scaled Autoregressive Projection Steering (SAPS): Token-level Steering to Induce Sarcasm and Humblebrag in Text** — Findings of EMNLP 2026
   **Sharath Naganna**, Ritesh Sur Chowdhury, Dipankar Srirag, Aditya Joshi.
2. **Are Arithmetic Heuristic Neurons Form-Invariant? A Mechanistic Analysis of Symbols, Text, and Code in LLMs** — BlackboxNLP @ EMNLP 2026
   **Sharath Naganna**, Tanvir Ahmed Sijan, Uddipta Kalita. [Paper] [Code]
3. **SAVIOR: Sample-efficient Adaptation of Vision-Language Models for OCR Representation** — VisionDocs @ WACV 2026
   Akshata A. Bhat, **Sharath Naganna**, Saiful Haq, Prashant Khatri, Krishna Chaitanya Reddy Tamataam, Neha Arun, Niyati Chhaya, Pushpak Bhattacharyya.
4. **"My life is miserable, have to sign 500 autographs everyday": Exposing Humblebragging, the Brags in Disguise** — Findings of ACL 2025
   **Sharath Naganna**, Saprativa Bhattacharjee, Biplab Banerjee, Pushpak Bhattacharyya.

## Research Experience

**Form-Invariance of Arithmetic Circuits in LLMs** — 2026
*Independent collaboration* | *Self-directed*

- Investigated whether arithmetic heuristic neurons are form-invariant across symbolic arithmetic, natural-language word problems, and Python code in three Llama-3 models.
- Localized these neurons with a two-stage pipeline combining attribution patching and activation patching, applied separately to each input format.
- Found a compact shared neuron set across all three formats, with targeted interventions confirming it is both necessary and sufficient for late-layer arithmetic computation.

**Mechanistic Connection between Zero-Shot and In-Context Learning** — 2026 – present
*IIT Kanpur* | *Dr. Ashutosh Modi*

- Compared sparse circuits underlying zero-shot and in-context learning using edge attribution patching with integrated gradients, across six tasks and three model families (Llama-3.1-8B, Mistral-7B, Qwen2.5-7B).
- Found asymmetric transfer: ICL circuits (median 5.95× larger) preserve zero-shot behaviour, while zero-shot circuits fail to recover ICL behaviour.
- Established that a median 86% of zero-shot circuit edges lie within the corresponding ICL circuit (3.59× above chance), evidence that ICL extends rather than replaces the zero-shot computation.

**Scaled Autoregressive Projection Steering for Figurative Language** — 2025 – present
*IIT Bombay* | *Dr. Aditya Joshi (UNSW)*

- Proposed SAPS, a residual-stream intervention that recomputes the steering coefficient at every decoding step so the intervention tracks incongruity as it unfolds, rather than fixing it at the first token as prior methods do.
- Analysed steering dynamics through layer-wise activation studies and token-level distribution divergence, improving grammaticality over fixed-coefficient baselines while preserving source content.

**Multilingual Visual Document Reranking** 2025 – present

*Flipkart–IITB collaboration | Dr. Soumen Chakrabarti (IITB), Abhinav Anand (Flipkart)*

- Built a multilingual dataset for visual document reranking: 250K queries across 6 languages with hard negatives mined from a multi-vector retriever (public release forthcoming).
- Designed the data pipeline, including a difficulty curriculum that cuts rank-1 queries from 69% to 21%, and NLLB-200 translation with round-trip quality checks.
- Trained a 1B reranker that outperforms the 2B DocReRank across all 8 UniDoc-Bench domains at half the size, benchmarked against 5 rerankers on ViDoRe v3 and UniDoc-Bench.

**Sample-efficient Adaptation of Vision-Language Models for OCR** Summer 2025, Summer 2026

*Hyperbots, Bengaluru | Dr. Niyati Chhaya*

- Proposed SAVIOR, a sample-efficient data curation strategy for adapting vision-language models to financial OCR, with a 2,234-document train set and a 509-document expert-annotated benchmark targeting vertical text, fine print and complex layouts.
- Introduced PaIRS, a structure-aware metric measuring pairwise spatial relationships between tokens; fine-tuned Qwen2.5-VL with LoRA, reaching 0.93 word recall and 0.89 F1 on downstream document QA.

## Industry Experience

**Software Engineer II** Jan 2022 – Jul 2024

*Cisco, Bengaluru*

- Progressed from Technical Intern to Software Engineer II in the Multi-Cloud Management team.
- Built backend services in Go and ETL pipelines between MongoDB and MySQL, with AWS, Azure and GCP account integration.

## Technical Skills

**Languages** C, Python, Golang, SQL

**Tools** PyTorch, Transformers, TransformerLens, vLLM, Flask

## Awards and Achievements

- Selected for the Heidelberg Laureate Forum, 2026.
- Travel grants: BlackboxNLP 2026 (workshop award); ACL 2025 (department award).
- All India Rank 325 of 123,967, Graduate Aptitude Test in Engineering (GATE) in Computer Science, 2025.
- Best Bachelor's Thesis (Research Category), JSS Science and Technology University; published at IEEE ICRASET 2024.

## Service

- Volunteer, AACL-IJCNLP 2025, IIT Bombay.